

Solutions

1. **Solution:** $P = A + \frac{3}{5}(B - A) = (-3, 5) + \frac{3}{5}(12, -12) = (-3, 5) + (7.2, -7.2) = (4.2, -2.2) = \left(\frac{21}{5}, -\frac{11}{5}\right).$

2. **Solution:** $P = \left(\frac{2 \cdot 8 + 3 \cdot 2}{5}, \frac{2 \cdot 5 + 3 \cdot (-1)}{5}\right) = \left(\frac{22}{5}, \frac{7}{5}\right).$

3. **Solution:** $M = \left(\frac{-6+10}{2}, \frac{4+(-2)}{2}\right) = (2, 1).$ $CD = \sqrt{(10+6)^2 + (-2-4)^2} = \sqrt{16^2 + (-6)^2} = \sqrt{256 + 36} = \sqrt{292} = 2\sqrt{73}.$

4. **Solution:** Midpoint equation: $P = \left(\frac{7+x_B}{2}, \frac{-3+y_B}{2}\right) = (2, 5).$ So $7 + x_B = 4 \Rightarrow x_B = -3$; $-3 + y_B = 10 \Rightarrow y_B = 13$; $B(-3, 13).$

5. **Solution:** Center $O =$ midpoint of E, F : $O = \left(\frac{-4+6}{2}, \frac{2+8}{2}\right) = (1, 5).$ $r = \frac{EF}{2} = \frac{1}{2}\sqrt{(6+4)^2 + (8-2)^2} = \frac{1}{2}\sqrt{10^2 + 6^2} = \frac{\sqrt{136}}{2} = \sqrt{34}.$ $C = 2\pi r = 2\pi\sqrt{34},$ $A = \pi r^2 = 34\pi.$

6. **Solution:** Diagonal $S_1S_2 = \sqrt{(5-1)^2 + (5-1)^2} = \sqrt{32} = 4\sqrt{2}.$ Square side $s = \frac{4\sqrt{2}}{\sqrt{2}} = 4.$ Perimeter = 16, area = 16.

7. **Solution:** Vector $\overrightarrow{AB} = (6, 0), \overrightarrow{BC} = (0, -3),$ so $\overrightarrow{CD} = \overrightarrow{BC} = (-6, 0)$ and $D = A + \overrightarrow{AD} = A + (0, -3) = (1, -1).$ Area = $\ell w = 6 \cdot 3 = 18$; perimeter = $2(6+3) = 18.$

8. **Solution:** $PQ = \sqrt{(4-0)^2 + (3-0)^2} = 5$, $PR = \sqrt{(-3-0)^2 + (4-0)^2} = 5$, $QR = \sqrt{(-3-4)^2 + (4-3)^2} = \sqrt{49+1} = \sqrt{50}$. Two equal sides $\Rightarrow$ isosceles. Also $5^2 + 5^2 = 50 = QR^2$, so it is *right isosceles*. Perimeter $= 5 + 5 + \sqrt{50} = 10 + 5\sqrt{2}$.

9. **Solution:** $x + (x + 18) = 90 \Rightarrow 2x = 72 \Rightarrow x = 36^\circ$, other $= 54^\circ$.

10. **Solution:** Linear pair $\Rightarrow$ sum 180° : $(3x + 15) + (5x - 3) = 180 \Rightarrow 8x + 12 = 180 \Rightarrow x = 21$. Angles: 78° and 102° .

11. **Solution:** Vertical $\Rightarrow$ equal: $2y + 14 = 5y - 13 \Rightarrow 3y = 27 \Rightarrow y = 9$. Angle $= 32^\circ$ (both).

12. **Solution:** Bisector $\Rightarrow$ equal halves: $x + 20 = 2x - 10 \Rightarrow x = 30$. Each half $= 50^\circ$, so $\angle ABC = 100^\circ$.

13. **Solution:** $M = \left(\frac{-2+x_Y}{2}, \frac{3+y_Y}{2}\right) = (4, -1)$. So $-2 + x_Y = 8 \Rightarrow x_Y = 10$; $3 + y_Y = -2 \Rightarrow y_Y = -5$. Hence $Y(10, -5)$.

14. **Solution:** Base $AB = 8$ (horizontal). Height is vertical distance from C to line $y = 0$, i.e. 6. Area $= \frac{1}{2}(8)(6) = 24$. $AC = \sqrt{(5-0)^2 + (6-0)^2} = \sqrt{25+36} = \sqrt{61}$.

15. **Solution:** Diagonals: $RT = \sqrt{(10-2)^2 + (7-1)^2} = \sqrt{64+36} = 10$; $SU = \sqrt{(2-10)^2 + (7-1)^2} = 10$ (equal). Area $= (10-2)(7-1) = 8 \cdot 6 = 48$; Perimeter $= 2(8+6) = 28$.

16. **Solution:** $\overrightarrow{AP} = (4 - (-2), 2 - 4) = (6, -2)$, $\overrightarrow{PB} = (13 - 4, -1 - 2) = (9, -3)$. Since $\overrightarrow{AP}$ and $\overrightarrow{PB}$ are collinear with same direction and $|\overrightarrow{PB}| = \frac{3}{2}|\overrightarrow{AP}|$, ratio $AP : PB = 2 : 3$. (Equivalently, compare component ratios $6 : 9 = 2 : 3$ and $-2 : -3 = 2 : 3$.)

17. **Solution:** General: $P = A + \frac{2}{5}(B - A) = \left(a + \frac{2}{5}(c - a), b + \frac{2}{5}(d - b)\right) = \left(\frac{3a+2c}{5}, \frac{3b+2d}{5}\right)$.
For $A(-1, 4)$, $B(9, -6)$: $P = \left(\frac{3(-1)+2(9)}{5}, \frac{3(4)+2(-6)}{5}\right) = \left(\frac{15}{5}, \frac{0}{5}\right) = (3, 0)$.

18. **Solution:** Slopes: $m_{QP} = \frac{0-2}{3-(-1)} = \frac{-2}{4} = -\frac{1}{2}$, $m_{QR} = \frac{-2-0}{7-3} = \frac{-2}{4} = -\frac{1}{2}$. Vectors $\overrightarrow{QP} = (-4, 2)$ and $\overrightarrow{QR} = (4, -2)$ are opposites; thus P, Q, R are collinear (angle Q is 180° , not right). Using distances: $QP = QR = \sqrt{(-4)^2 + 2^2} = \sqrt{20}$ and $PR = 2\sqrt{20}$, confirming collinearity. So the angle at Q is a straight angle.

19. **Solution:** Let sides be integers $\ell \geq w$, $2(\ell + w) = 38 \Rightarrow \ell + w = 19$. Diagonal $\sqrt{\ell^2 + w^2} = 13 \Rightarrow \ell^2 + w^2 = 169$. Also $(\ell + w)^2 = \ell^2 + w^2 + 2\ell w \Rightarrow 19^2 = 169 + 2\ell w \Rightarrow 361 = 169 + 2\ell w \Rightarrow \ell w = 96$. Solve $\ell + w = 19$, $\ell w = 96$: quadratic $t^2 - 19t + 96 = 0 \Rightarrow (t - 12)(t - 7) = 0$. Hence $(\ell, w) = (12, 7)$.

20. **Solution:** A square inscribed in a circle of radius r has diagonal $= 2r$. So side $s = \frac{2r}{\sqrt{2}} = r\sqrt{2}$. With $r = 5$, $s = 5\sqrt{2}$, area $= s^2 = 50$.